The High-Redshift Accretion Atlas

Reproducible Catalogue and Class-Aware Growth Diagnostics, Release v7.5

Nicole Jiang

27 August 2026

Abstract

We present v7.5 of a provenance-aware atlas of reported accreting black holes at $z \geq 4$. The release contains 234 literature measurements representing 219 physical objects in 218 host systems. Its science layer uses 209 measurements and 196 preferred physical-object rows with canonical numeric black-hole masses; 23 catalogue objects without such masses remain explicit exclusions. The atlas keeps heterogeneous evidence classes and mass methods separate, retains alternate measurements, and treats global ranks as navigation rather than demographic inference. A full-source audit corrects the Scholtz et al. JADES narrow/high-ionization subset to 21 objects at $z \geq 4$ and documents the paper/table count discrepancy without altering frozen v7.4 artifacts. Under the reference assumptions $z_{\mathrm{seed}} = 30$, $\epsilon = 0.1$, and no merger boost, luminous quasars occupy the highest growth-pressure region. These diagnostics measure pressure under stated assumptions; they do not identify a unique seed or accretion history.

1 Purpose and interpretation boundary

The atlas asks which reported objects place the strongest pressure on simple black-hole growth histories, and which measurements most affect that conclusion. It does not estimate a population mass function from the pooled catalogue. The input literature spans broad-line AGN selected by several JWST surveys [2, 3, 4, 5, 6, 7], luminous-quasar comparison samples [10, 12], host-selected broad-line candidates [9], an X-ray evidence history [13, 15], and JADES narrow/high-ionization candidates [16]. Selection functions, evidence strength, and mass estimators are therefore retained as analysis strata.

2 Catalogue construction and provenance

Every measurement has a stable measurement ID, physical-object ID, and host system ID. Source-native values, adopted canonical values, uncertainty kinds, method tags, URLs, available archive checksums, and identity decisions are preserved. Missing historical provenance is explicit: the 23 frozen-v1 Juodžbalis/JADES rows predate archive hashing and retain a documented unavailable checksum and extraction-date limitation rather than a fabricated value. Auxiliary coordinate and context inputs are attributed separately to the E-XQR-30 survey catalogue [11], the THRILS programme catalogue [8], the UHZ1 spectroscopy paper [14], and the versioned JADES DR3 GOODS-S prism catalogue. Alternate measurements are never silently averaged or deleted. Exactly one preferred measurement controls each object-level status in v7.5, while the complete history of measurement-level evidence statuses and bases remains in the object table. This resolves the case in which a weaker alternate candidate row had conservatively downgraded an otherwise secure preferred measurement.

The Scholtz et al. source audit deserves specific note. The paper reports 42 objects, while its source-native sample table contains 41 rows. Parsing that full table gives exactly 21 rows at $z \geq 4$. Frozen v7.4 contained

20 of them; v7.5 adds the omitted JADES-NS-GS00099671 row ($z = 5.936$) with its published host and bolometric observables, but no invented black-hole mass. The 21 candidates include one identity-linked broad-line object, so the current physical-object catalogue contains 20 narrow-line candidates from this source.

Table 1: Current v7.5 release scale and explicit analysis subsets.

Product or subset	Rows / objects
Measurements / physical objects / host systems	234 / 219 / 218
Source-local observables	1,106
Growth-eligible measurements / objects	209 / 196
Primary measurements / objects	182 / 171
Broad-line / luminous-quasar eligible objects	112 / 84
Catalogue-only physical objects	23
Alternate-measurement sensitivity pairs	13
Per-object growth images / compiled class grids	588 / 6

3 Growth model and uncertainty

The reference model follows exponential radiatively efficient growth [1],

$$M_{\rm BH}(t_{\rm obs}) = M_{\rm seed} \exp\left[f_{\rm Edd} \left(\frac{1-\epsilon}{\epsilon} \right) \frac{t(z_{\rm obs}) - t(z_{\rm seed})}{t_{\rm Edd}} \right] B_{\rm merge}, \tag{1}$$

with $t_{\rm Edd} \simeq 0.45$ Gyr. The main tables invert this relation to report required lifetime-average $f_{\rm Edd}$ for fixed seed masses and required $M_{\rm seed}$ for fixed $f_{\rm Edd}$. The baseline uses $z_{\rm seed} = 30$, $\epsilon = 0.1$, and $B_{\rm merge} = 1$.

Reported asymmetric statistical uncertainties in $\log_{10} M_{\rm BH}$ are propagated with 10,000 deterministic, per-object split-side draws. Method-scale systematic envelopes remain separate from the statistical sampling. A reported current Eddington ratio is not substituted for the lifetime-average growth rate.

4 Current results

The global navigation view places luminous quasars at the highest reference growth pressure. J1148+5251 is first in the point and uncertainty-aware views, with median required $f_{\rm Edd} = 1.214$ for a $10^2\,M_\odot$ seed. SDSS J0100+2802 and PSO J231-20 follow. The top eight uncertainty-ranked objects have unit Monte Carlo probability, to the stored precision, of requiring $f_{\rm Edd} > 1$ under the same light-seed reference. These results express tension with that reference history, not a claim of literally continuous super-Eddington accretion.

Within-class and within-class–mass-method ranks are the supported comparison scopes. All outputs explicitly forbid pooled demographic inference without a selection-function and completeness model. The alternate-measurement table contains 13 comparisons; it shows how retained measurement choice changes $\log_{10} M_{\rm BH}$ and required $f_{\rm Edd}$ while leaving the default-release choice auditable.

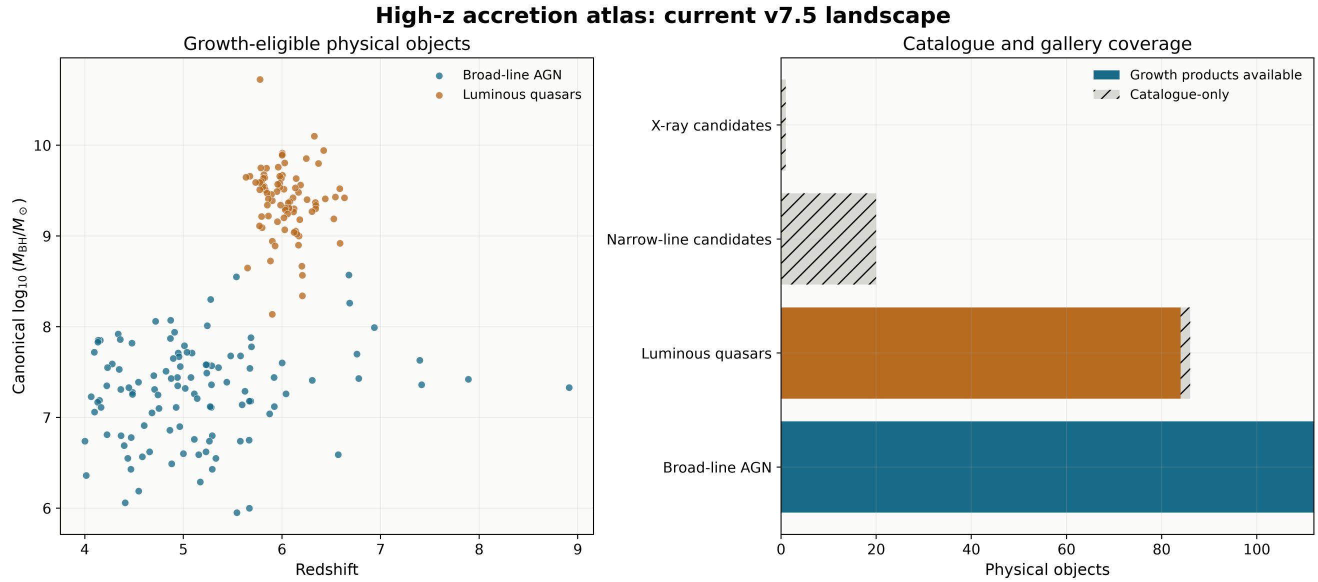

Figure 1: Current catalogue landscape. Left: all 196 growth-eligible physical objects in black-hole-mass–redshift space. Right: growth-product coverage for all 219 objects. The 23 catalogue-only objects remain available for provenance and evidence analyses but are not assigned numeric growth products.

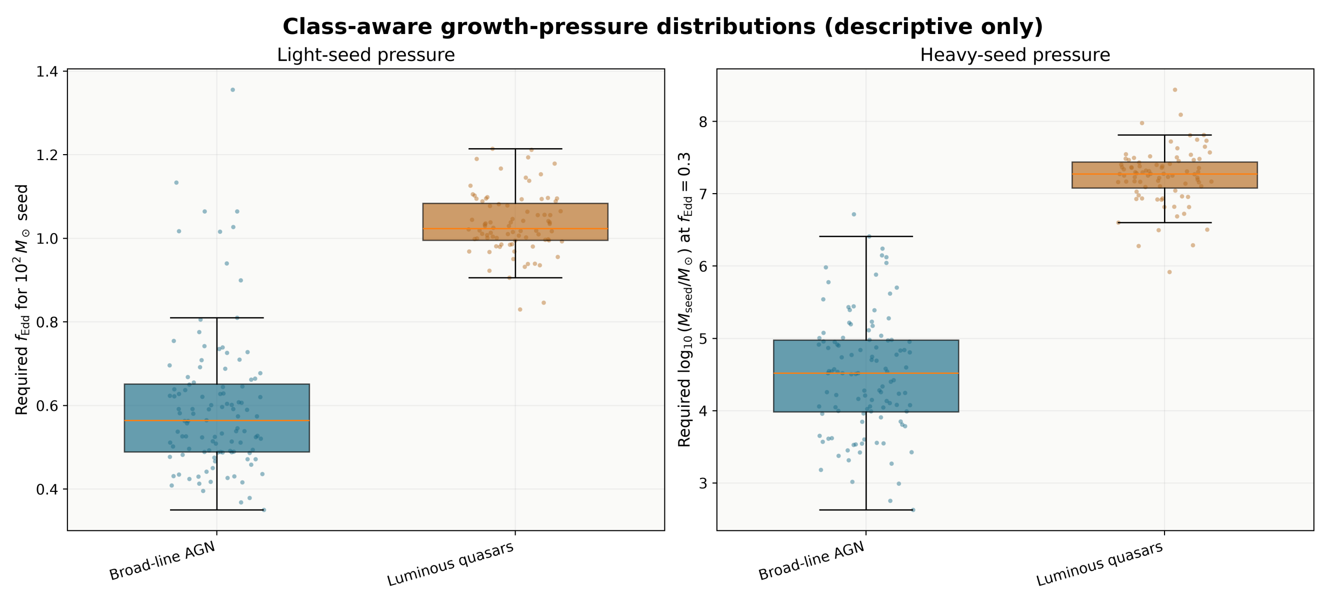

Figure 2: Descriptive class-aware growth-pressure distributions. The luminous quasar comparison sample and faint broad-line AGN are shown separately because their selection functions and mass regimes differ. Boxes and points summarize the released object-level table; they are not completeness-corrected demographic estimates.

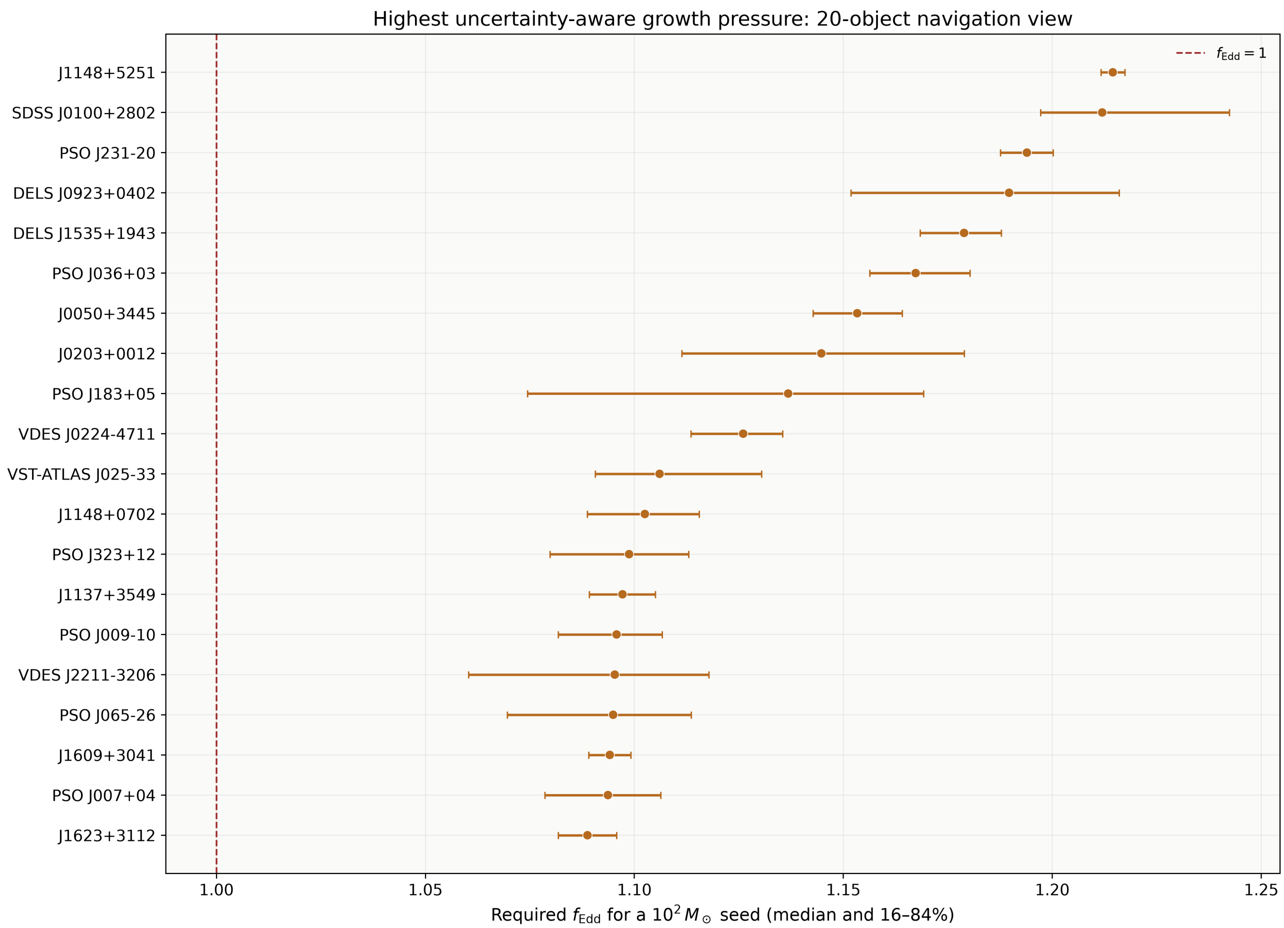

Figure 3: Twenty highest uncertainty-aware navigation entries. Points and bars show the median and 16th–84th percentiles of required f_{Edd} for a $10^2\,M_\odot$ seed. The dashed line marks $f_{\mathrm{Edd}} = 1$. Ranking is descriptive within the released assumptions.

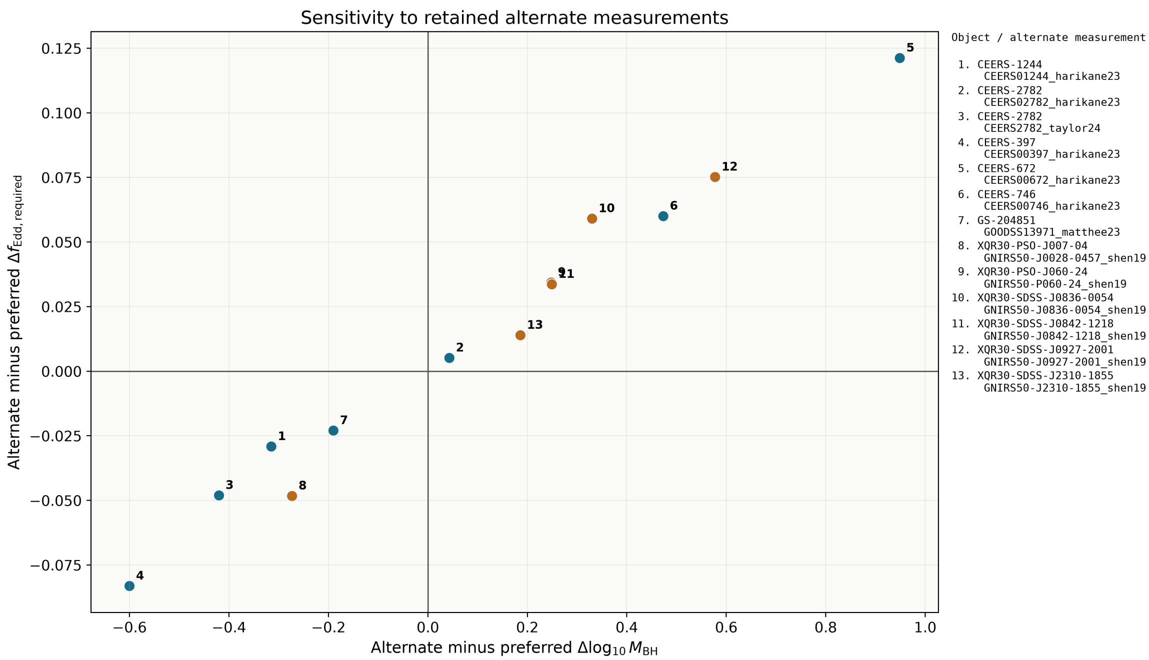

Figure 4: Sensitivity to 13 retained alternate measurements. Numbered points are keyed to stable object and measurement IDs, avoiding ambiguous overlapping labels. The plot quantifies release-choice sensitivity rather than treating alternates as independent objects.

5 Reproducibility and released artifacts

The Python 3.12 environment is exactly pinned. Catalogue, science, and figure manifests verify exact artifact membership and SHA-256 hashes. In-memory reproduction compares generated tables against checked-in CSVs without rewriting them. Regression tests enforce cardinalities, source anchors, identity links, evidence aggregation, uncertainty boundaries, ranking policy, figure resolution, and every inherited gallery path. Frozen releases remain independently verifiable.

The v7.5 figure layer provides four paper-facing figures. Its coverage table points to the byte-identical v7.4 per-object gallery because the 196 eligible objects and their growth inputs are unchanged. That gallery contains a parameter map, seed-redshift map, and growth track for each eligible object, plus six high-resolution class grids intended for close zooming. The 23 unavailable objects are recorded explicitly.

6 Limitations and next scientific work

- The catalogue is heterogeneous and cannot support pooled demographics without explicit selection and completeness models.
- Virial masses are method-dependent; available method systematics are reported separately and are not full posterior reconstructions.
- Narrow-line and X-ray candidates without canonical numeric masses are valuable evidence records but cannot enter the current growth ranking.
- The reference model compresses time-variable accretion, feedback, mergers, and radiative-state changes into effective parameters.
- Identity review uses deterministic candidates and documented decisions; crowded or imprecise cases may ultimately require probabilistic crossmatching.

The next scientific milestone is no longer basic catalogue assembly. It is a selection-aware inference layer: explicit survey selection functions, completeness modeling, method-calibrated hierarchical uncertainty, and an observing-priority matrix tied to which measurement would most change each object's physical interpretation.

7 Conclusion

v7.5 is a complete, internally consistent research release: the current catalogue, class-aware science tables, paper figures, full gallery coverage, provenance correction, evidence policy, and automated verification all refer to the same release. It provides reproducible observational triage across 219 objects while keeping unsupported demographic and formation-channel claims outside the scope of the data product.

References

[1] Dayal, P. 2024, *A&A*, 690, A182, doi:10.1051/0004-6361/202451481.

[2] Juodžbalis, I. et al. 2026, *MNRAS*, 546, stag086, doi:10.1093/mnras/stag086.

[3] Taylor, A. J. et al. 2025, *ApJ*, 986, 165, doi:10.3847/1538-4357/add15b.

[4] Matthee, J. et al. 2024, *ApJ*, 963, 129, doi:10.3847/1538-4357/ad2345.

[5] Lin, X. et al. 2024, *ApJ*, 974, 147, doi:10.3847/1538-4357/ad6565.

[6] Harikane, Y. et al. 2023, *ApJ*, 959, 39, doi:10.3847/1538-4357/ad029e.

[7] Davis, K. et al. 2026, arXiv:2602.23310v1.

[8] Hutchison, T. A. et al. 2025, THRILS redshift catalogue, arXiv:2512.12509v1.

[9] Ren, J. et al. 2025, *MNRAS*, 544, 211–233, doi:10.1093/mnras/staf1709.

[10] Mazzucchelli, C. et al. 2023, *A&A*, 676, A71, doi:10.1051/0004-6361/202346317.

[11] D'Odorico, V. et al. 2023, *MNRAS*, 523, 1399–1420, doi:10.1093/mnras/stad1468.

[12] Shen, Y. et al. 2019, *ApJ*, 873, 35, doi:10.3847/1538-4357/ab03d9.

[13] Bogdán, Á. et al. 2024, *Nature Astronomy*, 8, 126, doi:10.1038/s41550-023-02111-9.

[14] Goulding, A. D. et al. 2023, *ApJL*, 955, L24, doi:10.3847/2041-8213/acf7c5.

[15] Zou, F. et al. 2026, UHZ1 full-data X-ray reanalysis, arXiv:2603.24893v1.

[16] Scholtz, J. et al. 2025, *A&A*, 697, A175, doi:10.1051/0004-6361/202348804.